IT ASSET LIFECYCLE MANAGEMENT SYSTEM

Project Documentation

Developed Using Microsoft Power Platform

Project Name	IT Asset Lifecycle Management System
Platform	Microsoft Power Platform
Environment	Dev One
Primary Database	Microsoft Dataverse

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1. Project Overview

The IT Asset Lifecycle Management System is a centralized business application designed to simplify and automate the management of an organization's IT assets.

Organizations manage multiple IT resources, including laptops, desktops, monitors, printers, and mobile devices. Managing these assets manually can result in data duplication, inefficient asset allocation, missing records, delayed approvals, and difficulty tracking asset availability.

This project addresses these challenges by providing a centralized platform where employees can request IT assets and administrators can manage asset records, employee assignments, and approval processes.

The solution is developed using Microsoft Power Platform, integrating Power Apps, Microsoft Dataverse, Power Automate, Microsoft Teams, and Outlook.

2. Problem Statement

Many organizations still rely on spreadsheets, emails, and manual processes to manage IT assets. These approaches create several operational challenges:

- Difficulty maintaining accurate asset records.
- Lack of visibility into available and assigned assets.
- Manual tracking of employee-to-asset assignments.
- Delays in processing employee asset requests.
- Lack of a centralized approval mechanism.
- Difficulty monitoring asset maintenance and retirement.
- Limited reporting and visibility into asset utilization.

There is a need for a centralized and automated system that simplifies IT asset management and improves operational efficiency.

3. Project Objectives

The primary objectives of this project are:

1. Develop a centralized IT asset management platform.
2. Maintain accurate records of organizational IT assets.
3. Manage employee information and asset assignments.
4. Allow employees to submit IT asset requests.
5. Implement an automated approval and rejection workflow.
6. Notify administrators about new asset requests.
7. Maintain request status and approval history.
8. Improve asset visibility and resource utilization.
9. Reduce manual administrative work.

4. Proposed Solution

The proposed solution consists of two Power Apps applications supported by Dataverse and Power Automate.

Employee Application – Canvas App

Employees can:

- Access the application.
- View asset information.
- Submit new asset requests.
- Provide business justification for requests.
- Track request information and status.

Administrator Application – Model-Driven App

IT administrators can:

- View and manage IT assets.
- Add and update employee records.
- Review incoming asset requests.
- Monitor asset allocation and availability.
- Manage asset information throughout its lifecycle.

Automation – Power Automate

The automation layer handles:

- New asset request notifications.
- Approval and rejection processes.
- Automatic updates to request status.
- Microsoft Teams and Outlook notifications.

5. Technology Stack

Technology	Purpose
Microsoft Power Apps – Canvas	Employee-facing application
Microsoft Power Apps – Model-Driven	Administrator application
Microsoft Dataverse	Centralized database
Power Automate	Workflow and approval automation
Microsoft Teams	Real-time notifications
Microsoft Outlook	Email notifications and approval communication

Technology	Purpose
Power BI	Optional reporting and analytics integration

6. System Architecture

The system follows a centralized architecture in which both Power Apps applications use Microsoft Dataverse as their common data source.

Architecture Flow

Actor / Component	Flow
Employee	Employee -> Canvas App -> Dataverse -> Power Automate -> Teams / Outlook
IT Administrator	IT Administrator -> Model-Driven App -> Dataverse
Power BI (optional)	Dataverse data -> Reports and analytics

The centralized database ensures that information entered through one application can be accessed by authorized users through the other application.

7. Database Design

The project uses Microsoft Dataverse to store and manage application data.

7.1 IT Assets Table

This table stores information about all IT assets owned by the organization.

Column	Description
Asset Name	Name of the asset
Asset Tag	Unique asset identification tag
Category	Type of IT asset
Serial Number	Manufacturer serial number
Purchase Date	Date of purchase
Warranty Expiry	Warranty expiration date
Cost	Purchase cost
Status	Current asset status
Assigned To	Employee assigned to the asset

Asset categories: Laptop, Desktop, Monitor, Mobile Phone, Printer, Other.

Asset statuses: Available, Assigned, Under Maintenance, Retired.

7.2 Employees Table

This table maintains employee information.

Column	Description
Employee Name	Employee's full name
Employee ID	Unique employee identification
Department	Employee's department
Email	Official employee email address

7.3 Asset Requests Table

This table stores requests submitted by employees.

Column	Description
Request Title	Title of the request
Employee	Employee submitting the request
Asset Category	Requested asset type
Request Date	Date of submission
Business Justification	Reason for requesting the asset
Request Status	Current request status

Request status values: Pending, Approved, Rejected.

7.4 Relationships

The database contains relationships between the three primary tables:

- One employee can submit multiple asset requests.
- One employee can be assigned multiple IT assets.
- Each asset request is associated with an employee.
- Each asset can be assigned to an employee through the Assigned To lookup.

These relationships help maintain data consistency and reduce duplication.

8. Application Modules

8.1 Canvas Application – IT Asset Manager

The Canvas App provides an employee-friendly interface. Main functionalities:

- Home screen with asset overview.
- View Assets screen to display asset records.
- Add New Asset screen for authorized asset entry.
- Make Asset Request screen for submitting requests.
- Employee welcome message after successful email matching.

8.2 Model-Driven Application – IT Asset Administrator

The Model-Driven App provides an administrative interface. Main functionalities:

- IT Assets management.
- Employee records management.
- Asset Requests management.
- Asset status monitoring.
- Request review and administration.
- Charts for visualizing asset and request information.

8.3 Power Automate – Workflow Automation

Power Automate is used to reduce manual work and automate business processes.

Asset Request Approval Workflow

1. An employee submits an asset request.
2. A new record is created in Dataverse.
3. Power Automate detects the new record.
4. An approval request is sent to the administrator.
5. The administrator selects Approve or Reject.
6. The flow checks the approval outcome.
7. Dataverse updates the Request Status.
8. Notifications are sent through the configured communication channels.

Approval Outcomes

- **Approved:** Request Status is updated to Approved.
- **Rejected:** Request Status is updated to Rejected.

Additional automation can notify administrators when new employees are added.

9. Business Use Cases

The application can be used in:

- IT departments of corporate organizations.
- Educational institutions and universities.
- Software development companies.
- Small and medium-sized businesses.
- Organizations managing large numbers of IT devices.
- Companies with multiple departments and employee locations.

Example Business Scenario

An employee requires a new laptop for project-related work.

Instead of sending multiple emails to the IT department, the employee opens the Canvas App and submits an asset request with the required category and business justification.

The IT administrator receives the request and reviews it through the approval workflow. Once the administrator approves or rejects the request, the request status is updated in Dataverse.

This creates a structured and traceable asset request process.

10. Benefits of the Project

- Centralized asset information.
- Improved asset allocation and tracking.
- Reduced manual paperwork.
- Faster request processing.
- Better communication between employees and IT administrators.
- Improved data accuracy.
- Increased operational transparency.
- Better visibility into asset availability.
- Scalable application architecture.

11. Security and Access Management

The application uses Microsoft Power Platform and Dataverse security capabilities. Security considerations include:

- Role-based access for employees and administrators.
- Appropriate Dataverse table permissions.
- Restricted access to administrative functions.
- Secure authentication through Microsoft organizational accounts.
- Controlled access to sensitive employee and asset information.

> ***Note:*** *The email-matching login screen is a demonstration feature, not a secure authentication mechanism. Production deployment should use Microsoft Entra ID authentication and Dataverse security roles.*

12. Testing and Validation

The following test cases should be performed before deployment.

Test Case	Expected Result
Add a new IT asset	Asset record is created
View asset records	Available records are displayed

Test Case	Expected Result
Submit an asset request	New request appears in Dataverse
Submit request with valid employee	Employee is associated with request
Approve a request	Request Status becomes Approved
Reject a request	Request Status becomes Rejected
Add a new employee	Employee record is created
Trigger notification flow	Configured notification is delivered

Testing should be performed through both Canvas and Model-Driven Apps.

13. Limitations

- The application depends on internet connectivity and Microsoft Power Platform services.
- Power Platform licensing and organizational policies may affect connector availability.
- Notification delivery depends on Teams and Outlook connector configuration.
- The current solution is designed as a foundational asset management system and may require additional functionality for enterprise-scale deployment.
- Asset procurement, depreciation, and advanced maintenance scheduling are not included in the initial implementation.

14. Future Enhancements

The system can be extended with the following features:

1. QR code or barcode-based asset identification.
2. Automated warranty expiration reminders.
3. Asset maintenance scheduling.
4. Asset return and reassignment workflows.
5. Advanced Power BI dashboards.
6. Multi-location asset tracking.
7. Automated employee onboarding and offboarding.
8. Asset depreciation and cost analysis.
9. Integration with procurement systems.
10. Audit logs and detailed asset lifecycle history.

15. Learning Outcomes

Through this project, the following concepts and skills can be developed:

- Building Canvas Apps using Power Apps.

- Creating Model-Driven Apps.
- Designing relational data structures in Dataverse.
- Creating tables, columns, choices, and relationships.
- Implementing Power Fx formulas.
- Building automated workflows using Power Automate.
- Integrating Microsoft Teams and Outlook.
- Implementing approval and rejection processes.
- Understanding business application architecture.
- Testing and validating low-code applications.

16. Conclusion

The IT Asset Lifecycle Management System demonstrates how Microsoft Power Platform can be used to develop practical business applications and automate organizational processes.

By combining Canvas Apps, Model-Driven Apps, Dataverse, and Power Automate, the project provides a centralized solution for managing IT assets, employee information, and asset requests.

The solution reduces dependency on manual processes, improves communication, and creates a structured approach to asset management.

The project also establishes a foundation for future enhancements such as advanced reporting, maintenance automation, QR-based tracking, and enterprise-level asset management.